C S Vineeth

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PROFILE

AI Researcher and ML Engineer with experience building machine learning systems across fraud detection, cybersecurity analytics, computer vision, genomic variant classification, and IoT anomaly detection. Skilled in Python, SQL, LightGBM, XGBoost, PyTorch, TensorFlow, YOLO, GNNs, transformers, and AWS-based deployment. Strong focus on leakage-safe validation, explainability, model benchmarking, and production-ready inference workflows.

SKILLS

Programming & Data: Python, SQL, Pandas, NumPy
Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, feature selection, SHAP
Deep Learning: PyTorch, TensorFlow, CNNs, YOLO, LSTM, Transformers, Autoencoders, GNNs
ML Systems: Batch inference, real-time inference APIs, model benchmarking, pipeline automation
Cloud & Deployment: AWS EC2, S3, IAM, Vercel, TorchScript, ONNX, API integration
Evaluation: ROC-AUC, PR-AUC, F1, mAP, confusion matrix, cross-validation, threshold tuning

EXPERIENCE

CBK Infotech India Pvt. Ltd. - AI Researcher
Bengaluru, India | Apr 2023 - Present

Fraud Detection & Risk Intelligence

- Built fraud-risk scoring pipelines across large-scale transactional and relational datasets using LightGBM, XGBoost, TabTransformer, and graph-based modelling.
- Achieved ROC-AUC 0.9995 and F1 0.986 on held-out evaluation across 500K+ records after class-imbalance handling, feature engineering, and threshold optimization.
- Benchmarked classical ML, transformer-based tabular models, and GNN variants to validate transactional and relational fraud signals.
- Improved relational fraud detection accuracy by 2.3% through graph aggregation strategy tuning and feature construction.
- Delivered reproducible batch inference and evaluation pipelines for model benchmarking, risk scoring, and decision-support analysis.

Healthcare Computer Vision

- Developed YOLO-based object detection pipelines for dental image analysis, including preprocessing, training, test-set validation, and visual result inspection.
- Achieved Precision 0.9843, Recall 0.9820, and mAP@50 0.9861 on test-set dental object detection evaluation.
- Built multi-stage medical imaging workflows covering detection, embedding-based similarity matching, and evaluation reporting.
- Automated preprocessing, validation, and model evaluation workflows, reducing repeated manual experimentation effort.

Genomic Variant Classification Research

- Built a ClinVar preprocessing pipeline from 8.9M raw variant records, applying review-status filtering, conflict resolution, SNV selection, duplicate removal, and allele validity checks.
- Created a cleaned binary SNV classification dataset of 637K+ variants with pathogenic/benign labels and class-imbalance analysis.
- Engineered 54 genomic features including nucleotide encodings, transition/transversion signals, GC/CpG context, k-mer frequencies, gene-level frequency, positional features, and functional scores.
- Built model-ready feature matrices for downstream deep learning and ML benchmarking with null-value handling, imputation, scaling, and correlation analysis.

Education Analytics & Predictive Modelling

- Built ML pipelines over 10M+ behavioural interaction events to predict engagement and outcome-risk patterns.
- Improved model reliability through feature engineering, label refinement, class balancing, ensemble modelling, and cross-validation stability checks.

TCR Innovation - AI Research Intern
Mumbai, India | Nov 2022 - Jan 2023

- Developed computer vision models for object counting, defect detection, and segmentation, improving detection accuracy by 17% over baseline models.
- Built recommendation pipelines combining collaborative filtering, content-based similarity, and sentiment-derived features.
- Deployed inference workflows on AWS EC2/S3 with batching and caching, reducing latency by 35% and infrastructure cost by 22%.
- Created reusable preprocessing, training, and evaluation scripts for faster experimentation across multiple AI prototypes.

SELECTED PROJECTS

Cybersecurity Intrusion Detection | *Python, XGBoost, LightGBM, Random Forest, SHAP, Mutual Information*

- Built a network intrusion detection pipeline on 211K+ network-flow records with 44 raw columns, including identifier removal, categorical encoding, feature selection, and model validation.
- Selected top behavioural features using mutual information and validated feature influence using SHAP-based global importance.
- Benchmarked Random Forest, XGBoost, and LightGBM, achieving AUROC above 0.999 on stratified test evaluation.
- Performed feature-drop sanity testing and confusion-matrix error analysis to verify whether performance depended on only a few dominant signals.

IoT Sensor Anomaly Detection | *TensorFlow, LSTM Autoencoder, Transformer Autoencoder, Time-Series Modelling*

- Built unsupervised anomaly detection pipelines on 2.2M+ cleaned Intel Berkeley sensor readings using temperature, humidity, light, and voltage signals.
- Generated 221,976 sliding-window sequences using 50-timestep windows and leakage-safe train-only MinMax normalization.
- Compared LSTM and Transformer autoencoders for zero-day anomaly detection using reconstruction-error scoring, AUROC, PR-AUC, and adaptive thresholding.
- Simulated real-time streaming inference by scoring incoming sensor windows sequentially for edge-style anomaly detection.

LLM-Based Text Intelligence Platform | *LLM APIs, Embeddings, React/Next.js, Vercel (Click here)*
- Built and deployed a web-based LLM application for real-time text analysis and controlled content generation.
- Implemented prompt templates, embedding-based control, session caching, and API error handling to improve response consistency.
- Optimized request orchestration, achieving approximately 1.3s average response time during live testing.

AI Voice Assistant | *Speech-to-Text, NLP, Async APIs, Vercel (Click here)*
- Built a voice-driven AI assistant integrating speech-to-text, intent recognition, and contextual response generation.
- Implemented asynchronous inference flow and modular APIs to support low-latency, cross-device access.

Recommendation & Sentiment Intelligence System | *Python, NLP, Collaborative Filtering, Content-Based Filtering*
- Built hybrid recommendation pipelines combining collaborative filtering, content-based similarity, and sentiment-derived features for personalized ranking.
- Processed user-item interaction signals, textual feedback, and similarity features to improve recommendation relevance across prototype use cases.
- Created reusable preprocessing, scoring, and evaluation workflows for faster experimentation across recommendation-system variants.

RESEARCH & ENGINEERING STRENGTHS
- Strong validation discipline across held-out testing, stratified splits, cross-validation, confusion-matrix analysis, threshold tuning, and leakage-safe preprocessing.
- Built reusable ML experiment workflows including preprocessing artifacts, saved feature matrices, model checkpoints, metric tables, visualizations, and reproducible evaluation scripts.
- Applied explainability and model-analysis methods including SHAP, mutual information, correlation analysis, feature-drop sanity checks, class-wise error review, and visual inspection.
- Developed deployment-oriented workflows using batch scoring, API inference, caching, AWS EC2/S3, Vercel deployment, TorchScript, and ONNX export.
- Converted notebook experiments into reviewer-ready documentation with dataset summaries, preprocessing notes, metric tables, plots, and reproducibility details.
- Maintained objective-to-execution mapping across datasets, preprocessing, modelling, validation, deployment readiness, and final reporting artifacts.

RESEARCH WORK
- Prepared applied AI research workflows across fraud detection, genomic variant classification, cybersecurity analytics, computer vision, and IoT anomaly detection.
- Documented dataset preparation, preprocessing decisions, model comparisons, validation protocols, evaluation metrics, and deployment-readiness checks for technical review.
- Focused on reproducible experimentation using saved artifacts, structured notebooks, visual result analysis, and clear mapping between objectives, methods, and reported outcomes.

EDUCATION
BMS Institute of Technology and Management, Bengaluru
Bachelor of Engineering, Artificial Intelligence and Machine Learning | 2019 – 2023

CERTIFICATIONS
- IBM - Machine Learning with Python, Honors
- Google - Preparing Data for Exploration
- Udemy - Python for Data Science

ACHIEVEMENTS
- Winner, BMSIT Project-Based Learning Open Day 2021
- Third Runner-Up, BMSIT Academic Project Showcase 2022